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FIXING ASSEMBLY AND ELECTRONIC DEVICE

Abstract

A fixing assembly is adapted to fix a storage device. The storage device includes a first side surface, two opposite second side surfaces, a first hole located on the first side surface and two second holes located on the two second side surfaces. The fixing assembly includes a supporting body with a base plate and a side plate and two moving components disposed on two sides of the base plate. The side plate has a first positioning column. An accommodating space is formed between the supporting body and the two moving components. The two moving components includes two second positioning columns. The first positioning column and the two second positioning columns protrude towards the accommodating space. When the storage device is disposed in the accommodating space, the first positioning column inserts into the first hole, and the two second positioning columns are adapted to insert into the two second holes.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims the priority benefit of Taiwan application serial no. 113106536, filed on Feb. 23, 2024. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND
[0002] Technical Field
[0003] This disclosure relates to a fixing assembly and an electronic device, and in particular to a fixing assembly and an electronic device that may quickly fix a storage device.
Description of Related Art
[0004] Generally speaking, storage hard disks in servers or industrial computers are installed in servers or industrial computers by means of a fixed bracket to which the storage hard disks are screwed. However, such a design results in higher labor hours for assembly and material preparation, requires more equipment for the production process, and may cause problems such as screws falling out during the assembly process. In addition, when a storage hard drive needs to be serviced, removing and reattaching the screws can increase the time required for servicing. Therefore, how quickly storage hard disks can be fixed or removed from the fixed brackets is a topic of discussion in this field.
SUMMARY
[0005] The disclosure provides a fixing assembly and an electronic device, capable of quickly fixing a storage device.
[0006] The fixing assembly of the disclosure is adapted to fix a storage device. The storage device includes a first side surface, two second side surfaces opposite to each other and adjacent to the first side surface, a first hole located on the first side, and two second holes located on the two second side surfaces respectively. The fixing assembly includes a supporting body and two moving components. The supporting body includes a base plate and a side plate connected to each other. The base plate is adapted to carry the storage device. The side plate has a first positioning column extending in a first axial direction. The two moving components are movably disposed on two sides of the base plate in a second axial direction. An accommodating space is formed between the supporting body and the two moving components. The two moving components respectively include two second positioning columns extending in the second axial direction. The first positioning column and the two second positioning columns protrude towards the accommodating space. When the storage device is disposed in the accommodating space, the first positioning column is inserted into the first hole, and the two second positioning columns are adapted to be inserted into the two second holes.
[0007] An electronic device of the disclosure includes a storage device and a fixing assembly. The storage device includes a first side surface and two second side surfaces opposite to each other. The first side surface has a first hole. The two second side surfaces opposite to each other are adjacent to the first side surface respectively. Each of the second side surface has a second hole. The fixing assembly includes a supporting body and two moving components. The supporting body includes a base plate and a side plate connected to each other. The base plate is adapted to carry the storage device. The side plate has a first positioning column extending in a first axial direction. The two moving components are movably disposed on two sides of the base plate in a second axial direction. An accommodating space is formed between the supporting body and the two moving components. The two moving components respectively include two second positioning columns

extending in the second axial direction. The first positioning column and the two second positioning columns protrude towards the accommodating space. When the storage device is disposed in the accommodating space, the first positioning column is inserted into the first hole, and the two second positioning columns are adapted to be inserted into the two second holes.
[0008] Based on the above, the fixing assembly of the electronic device in the disclosure includes the supporting body and the moving component. The supporting body includes the first positioning column, and the moving component includes the second positioning column. The storage device may be fixed in the fixing assembly by inserting the first positioning column and the second positioning column into the first hole and the second hole of the storage device respectively. Since the moving component of the disclosure may be moved relative to the supporting body, a user may quickly fix the storage device or quickly dismantle the storage device by moving the two moving components close to or far away from the supporting body, so that the second positioning column may be inserted into or removed from the second hole of the storage device, and this design may enhance the efficiency of the assembly or maintenance.
[0009] To make the aforementioned more comprehensible, several embodiments accompanied with drawings are described in detail as follows.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0010] The accompanying drawings are included to provide a further understanding of the disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate example embodiments of the disclosure and, together with the description, serve to explain the principles of the disclosure.
[0011] FIG. **1** is a schematic diagram of an electronic device according to an embodiment of the disclosure.
[0012] FIG. **2** is a schematic diagram of a storage device fixed to a fixing assembly of FIG. **1**.
[0013] FIG. **3** is an exploded diagram of the storage device and the fixing assembly of FIG. **2**.
[0014] FIG. **4** is a schematic diagram of the storage device in FIG. **3** from another perspective.
[0015] FIG. **5** is a schematic diagram of the fixing assembly of FIG. **2**.
[0016] FIG. **6**A shows a moving component of FIG. **5** being pulled outward.
[0017] FIG. **6**B shows a storage device disposed in a fixing assembly of FIG. **6**A.
[0018] FIG. **6**C shows the storage device of FIG. **6**B moving in a first axial direction.
[0019] FIG. **6**D shows a moving component of FIG. **6**C moving in a second axial direction.
[0020] FIG. **7** is a schematic diagram of a fixing assembly according to another embodiment of the disclosure.
[0021] FIG. **8** is an exploded diagram of a storage device and the fixing assembly of FIG. **7**.
[0022] FIG. **9** shows two moving components of FIG. **7** being pulled outward.
[0023] FIG. **10** is a partially enlarged diagram of the moving component and a supporting body of FIG. **9** from another perspective.
DESCRIPTION OF THE EMBODIMENTS
[0024] FIG. **1** is a schematic diagram of an electronic device according to an embodiment of the disclosure. FIG. **2** is a schematic diagram of a storage device fixed to a fixing assembly of FIG. **1**. Referring to FIG. **1** and FIG. **2**, an electronic device **100** in this embodiment is, for example, a server or an industrial computer. The electronic device **100** includes multiple storage devices **110** and multiple fixing assemblies **120**. The storage device **110** is fixed to the fixing assembly **120** as shown in FIG. **2**, and is disposed in the electronic device **100** through the fixing assembly **120**.
[0025] FIG. **3** is an exploded diagram of the storage device and the fixing assembly of FIG. **2**. FIG. **4** is a schematic diagram of the storage device in FIG. **3** from another perspective. Referring to

FIG. **3** and FIG. **4**, the storage device **110** includes a first side surface **111** (shown in FIG. **4**), two second side surfaces **112** opposite to each other and adjacent to the first side surface **111**, two first holes **113** located on the first side surface **111** (as shown in FIG. **4**), and two second holes **114** respectively located on the two second side surfaces **112**. The fixing assembly **120** includes a supporting body **121**, two moving components **122**, and an appearance component **123**. The supporting body **121** includes a base plate **121***a* and a side plate **121***b* connected to each other. The base plate **121***a* is adapted to carry the storage device **110**. The side plate **121***b* includes two first positioning columns **1211** extending in a first axial direction A**1** and is connected to the appearance part **123**. The two moving components **122** are movably disposed on two sides of the base plate **121***a* in a second axial direction A**2**. An accommodating space S is formed between the supporting body **121** and the two moving components **122**, and the storage device **110** is located in the accommodating space S as shown in FIG. **2**. The two moving components **122** respectively include two second positioning columns **1221** extending in the second axial direction A**2**. The first positioning column **1211** and the two second positioning columns **1221** protrude toward the accommodating space S as shown in FIG. **3**.

[0026] As shown in FIG. **2**, when the storage device **110** is disposed in the accommodating space S of the fixing assembly **120**, the two first positioning columns **1211** are respectively inserted into the two first holes **113** of the storage device **110**, and the two second positioning columns **1221** are respectively adapted to be inserted into the two second holes **114** of the storage device **110**.

[0027] As mentioned above, the fixing assembly **120** of the electronic device **100** of this embodiment includes the supporting body **121** and the moving component **122**. The supporting body **121** includes the first positioning column **1211**, and the moving component **122** includes the second positioning column **1221**. The storage device **110** may be fixed in the fixing assembly **120** by inserting the first positioning column **1211** and the second positioning column **1221** into the first hole **113** and the second hole **114** of the storage device **110** respectively. Since the moving component **122** of the disclosure may be moved relative to the supporting body **121**, a user may quickly fix the storage device **110** or quickly dismantle the storage device **110** by moving the two moving components **122** close to or far away from the supporting body **110**, so that the second positioning column **1221** may be inserted into or removed from the second hole **114** of the storage device **110**, and this design may enhance the efficiency of the assembly or maintenance.

[0028] In this embodiment, the first axial direction A**1** is perpendicular to the second axial direction A**2**, but the disclosure is not limited thereto.

[0029] In this embodiment, the storage device **110** is, for example, an E3.S solid state drive, but the type of the storage device **110** is not limited in the disclosure. In addition, in this embodiment, a number of the first hole **113**, the second hole **114**, the first positioning column **1211**, and the second positioning column **1221** is two respectively, but the number of the first hole **113**, the second hole **114**, the first positioning column **1211**, and the second positioning column **1221** is not limited thereto.

[0030] The following describes in detail the structure and combination of the supporting body **121** and the moving component **122**.

[0031] FIG. **5** is a schematic diagram of the fixing assembly of FIG. **2**. Referring to FIG. **3** and FIG. **5**. The two moving components **122** partially overlap the supporting body **121**. The supporting body **121** includes a slider **1212**. In this embodiment, the slider **1212** may be a rivet or a screw, but the disclosure is not limited thereto. Each of the moving components **122** includes a limiting chute **1222** extending in the second axial direction A**2** (shown in FIG. **3**). The slider **1212** is slidably disposed on the limiting chute **1222** to limit a moving distance of the moving component **122** relative to the supporting body **121** in the second axial direction A**2**. In this embodiment, the supporting body **121** is riveted to the moving component **122** through the slider **1212**, for example, so that the supporting body **121** and the moving component **122** are not separated from each other in a direction perpendicular to a bottom wall **1224**.

[0032] In this embodiment, a number of the slider **1212** and the limiting chute **1222** is four respectively. The limiting chutes **1222** are located at opposite ends of the moving component **122** in the first axial direction A**1**, and the sliders **1212** are located at four corners of the supporting body **121**, but the disclosure does not limit the number and position of the limiting chute **1222** and the slider **1212**. On the other hand, in other embodiments, the relationship between the each of the moving components, the supporting body, the slider, and the limiting chute may also be that the each of the moving components includes the slider and the supporting body includes the limiting chute, and the disclosure is not limited thereto.

[0033] Referring to FIG. **5**, the each of the moving components **122** further includes a bottom wall **1224** and a side wall **1225** standing from the bottom wall **1224**. The bottom wall **1224** includes a holding groove **1224***a*, and the side wall **1225** includes a second positioning column **1221**. In this embodiment, when the user wants to move the each of the moving components **122** relative to the supporting body **121** in the second axial direction A**2**, the user may reach into the holding groove **1224***a* of the bottom wall **1224** with a finger to operate, so that the second positioning column **1221** may be inserted into or moved out of the second hole **114** of the storage device **110**, i.e., the holding groove **1224***a* improves the ease of operation of the user.

[0034] FIG. **6**A shows a moving component of FIG. **5** being pulled outward. Referring to FIG. **3**, FIG. **5**, and FIG. **6**A, the each of the moving components **122** includes multiple elastic pieces **1223** (shown in FIG. **3**), and the supporting body **121** includes multiple sets of two through holes **1213** arranged in the second axial direction A**2**. Each of the elastic pieces **1223** includes a convex hull **1223***a*, and the convex hull **1223***a* extends into one of the corresponding two through holes **1213**. Specifically, in this embodiment, when the moving component **122** is in a fixed position relative to the supporting body **121** as shown in FIG. **5**, the convex hull **1223***a* extends into an inner through hole **1213** of the two through holes **1213**. When the moving component **122** is in a pulled-out position relative to the supporting body **121** as shown in FIG. **6**A, the convex hull **1223***a* extends into an outer through hole **1213** of the two through holes **1213**. However, the disclosure is not limited thereto. In other embodiments, when the moving component is in the fixed position relative to the supporting body, the convex hull may also extend into the outer hole of the two holes. When the moving component is in the pulled-out position relative to the supporting body, the convex hull may also extend into the inner hole of the two holes.

[0035] In this embodiment, when the convex hull **1223***a* extends into the through hole **1213**, the elastic piece **1223** generates a sense of positioning feedback with the supporting body **121**, so that the user may confirm that the convex hull **1223***a* has indeed extended into the through hole **1213** through the sound and the sense of positioning feedback. In addition, in this embodiment, the fixing assembly **120** further supports the supporting body **121** by means of the elastic piece **1223** to ensure that the convex hull **1223***a* does not come loose from the through hole **1213**.

[0036] In this embodiment, a number of the elastic piece **1223** of the each of the moving components **122** is two, and a number of sets of the two through holes **1213** is four. However, the disclosure does not limit the number of the elastic piece **1223** and the number of sets of the two through holes **1213**.

[0037] The following specifies the steps for fixing the storage device **110** to the fixing assembly **120**.

[0038] FIG. **6**B shows a storage device disposed in a fixing assembly of FIG. **6**A. FIG. **6**C shows the storage device of FIG. **6**B moving in a first axial direction. FIG. **6**D shows a moving component of FIG. **6**C moving in a second axial direction. Referring to FIG. **6**A to FIG. **6**D, first, as shown in FIG. **6**A, the each of the moving components **122** is pulled away from the supporting body **121** in the second axial direction A**2**, so that the two moving components **122** are away from each other, and the convex hull **1223***a* extends from the inner through hole **1213** into the outer through hole **1213** of the two through holes **1213**. Next, as shown in FIG. **6**B, the storage device **110** is placed on the supporting body **121**, and the storage device **110** is moved closer to the appearance component **123** in the first axial direction A**1**, so that the first positioning column **1211** is inserted into the first hole **113** of the storage device **113** as shown in FIG. **6**C. Finally, the two moving components **122** are moved in the second axial direction A**2** toward the supporting body **121**, so that the two moving components **122** are close to each other, and the convex hull **1223***a* extends from the outer through hole **1213** into the inner through hole **1213** of the two through holes **1213**. When the second positioning column **1221** is inserted into the second hole **114** of the storage device **110** as shown in FIG. **6**D, the steps of fixing the storage device **110** to the fixing assembly **120** are completed.

[0039] FIG. **7** is a schematic diagram of a fixing assembly according to another embodiment of the disclosure. FIG. **8** is an exploded diagram of a storage device and the fixing assembly of FIG. **7**. Referring to FIG. **3**, FIG. **7**, and FIG. **8**, the difference between a fixing assembly **120′** of this embodiment and the fixing assembly **120** of FIG. **3** is that a slider **1212′** of this embodiment is formed by folding a supporting body **121′**. In addition, a moving component **122′** further includes a hollow area **1226′** and the supporting body **121′** further includes a heat dissipation hole **1216′**. The following is an explanation of the differences only.

[0040] Compared with FIG. **3** where the slider **1212** of the fixing assembly **120** is a rivet riveted to the supporting body **121**, in this embodiment, the slider **1212′** of the fixing assembly **120′** is formed by folding the supporting body **121′**, and is integrated with the supporting body **121′**. Accordingly, the assembly process and assembly time of the fixing assembly **120′** may be reduced, which in turn reduces the cost.

[0041] In other embodiments, the slider may also be formed by folding the moving component and is integrated with the moving component, and the supporting body includes a limiting chute, and the disclosure is not limited thereto.

[0042] The following describes in detail the structure and combination of the supporting body **121′** and the moving component **122′**.

[0043] FIG. **9** shows two moving components of FIG. **7** being pulled outward. FIG. **10** is a partially enlarged diagram of the moving component and a supporting body of FIG. **9** from another perspective. Referring to FIG. **8** to FIG. **10**, a base plate **121***a*′ of the supporting body **121′** includes a first surface **1214′** and a second surface **1215′** opposite to each other (shown in FIG. **10**). Each moving components **122′** includes two elastic pieces **1223′** and two limiting chutes **1222′**. The base plate **121***a*′ of the supporting body **121′** includes four sets of two through holes **1213′** and four sliders **1212′**. As shown in FIG. **10**, the elastic piece **1223′** is located next to the second surface **1215′** and abuts against the second surface **1215′**. The moving component **122′** rests on the first surface **1214′** at a part **1228′** enclosing the limiting chute **1222′** as shown in FIG. **9**. Specifically, the each moving component **122′** has a bend **1227′** between the elastic piece **1223′** and the limiting chute **1222′**, so that the elastic piece **1223′** is in a different plane from the part **1228′** of the moving component **122′** enclosing the limiting chute **1222′**.

[0044] In this embodiment, the fixing assembly **120′** respectively abuts the second surface **1215′** and the first surface **1214′** through the elastic piece **1223′** and the part **1228′** of the moving component **122′** enclosing the limiting chute **1222′**, so that the supporting body **121′** and the moving component **122′** are not separated from each other in a direction perpendicular to the first surface **1214′**.

[0045] Referring to FIG. **7**, Each moving component **122′** includes a bottom wall **1224′** and a side wall **1225′** standing from the bottom wall **1224′**. Multiple hollow areas **1226′** are located next to the bend **1227′** and located at a junction of the bottom wall **1224′** and the side wall **1225′**, so that the moving component **122′** can be bent between the elastic piece **1223′** and the limiting chute **1222′**.

[0046] In this embodiment, each moving component **122′** includes two hollow areas **1226′**, but a number of the hollow area **1226′** is not limited thereto.

[0047] Referring to FIG. **7**, the base plate **121***a*′ of the supporting body **121′** of this embodiment further includes multiple heat dissipation holes **1216′**, so that the storage device **110** has a good heat dissipation effect. In this embodiment, a number of the heat dissipation hole **1216′** is four, but the number of the heat dissipation hole **1216′** is not limited thereto.

[0048] To sum up, the fixing assembly of the electronic device in the disclosure includes the supporting body and the moving component. The supporting body includes the first positioning column, and the moving component includes the second positioning column. The storage device may be fixed in the fixing assembly by inserting the first positioning column and the second positioning column into the first hole and the second hole of the storage device respectively. Since the moving component of the disclosure may be moved relative to the supporting body, a user may quickly fix the storage device or quickly dismantle the storage device by moving the two moving components close to or far away from the supporting body, so that the second positioning column may be inserted into or removed from the second hole of the storage device, and this design may enhance the efficiency of the assembly or maintenance.

[0049] It will be apparent to those skilled in the art that various modifications and variations can be made to the disclosed embodiments without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the disclosure covers modifications and variations provided that they fall within the scope of the following claims and their equivalents.

Claims

1. A fixing assembly, adapted to fix a storage device, the storage device comprising a first side surface, two second side surfaces opposite to each other and adjacent to the first side surface, a first hole located on the first side surface, and two second holes located on the two second side surfaces respectively, wherein the fixing assembly comprising: a supporting body, comprising a base plate and a side plate connected to each other, wherein the base plate is adapted to carry the storage device, and the side plate has a first positioning column extending in a first axial direction; and two moving components, movably disposed on two sides of the base plate in a second axial direction, wherein an accommodating space is formed between the supporting body and the two moving components, the two moving components respectively comprise two second positioning columns extending in the second axial direction, and the first positioning column and the two second positioning columns protrude towards the accommodating space; when the storage device is disposed in the accommodating space, the first positioning column inserted into the first hole, and the two second positioning columns adapted to be inserted into the two second holes.

2. The fixing assembly according to claim 1, wherein the first axial direction is perpendicular to the second axial direction.

3. The fixing assembly according to claim 1, wherein the two moving components partially overlap the supporting body, one of each of the moving components and the supporting body comprises a limiting chute extending in the second axial direction, and the other one comprises a slider, the slider slidably disposed on the limiting chute.

4. The fixing assembly according to claim 3, wherein the slider is a rivet or a screw disposed on the corresponding moving component or supporting body.

5. The fixing assembly according to claim 3, wherein the slider is formed by folding the disposed moving component or supporting body, and is integrated with the disposed moving component or supporting body.

6. The fixing assembly according to claim 1, wherein one of each of the moving components and the supporting body comprises an elastic piece, and the other one comprises two through holes arranged in the second axial direction, the elastic piece comprises a convex hull, when the moving component is in a fixed position relative to the supporting body, the convex hull extends into one of the two through holes, when the moving component is in a pulled-out position relative to the supporting body, the convex hull extends into the other one of the two through holes.

7. The fixing assembly according to claim 6, wherein the base plate comprises a first surface and a second surface opposite to each other, the moving component comprises the elastic piece and a limiting chute, the base plate comprises the two through holes and a slider, the elastic piece is located next to the second surface and abuts against the second surface, and the moving component rests on the first surface at a part enclosing the limiting chute.

8. The fixing assembly according to claim 7, wherein the each of the moving components has a bend between the elastic piece and the limiting chute, such that the elastic piece is in a different plane from the part of the moving component enclosing the limiting chute, the each of the moving components comprises a bottom wall and a side wall standing from the bottom wall, and a hollow area is located next to the bend and located at a junction of the bottom wall and the side wall.

9. The fixing assembly according to claim 1, wherein each of the moving components comprises a bottom wall and a side wall standing from the bottom wall, the bottom wall comprises a holding groove, and the side wall comprises the second positioning column.

10. The fixing assembly according to claim 1, wherein the base plate comprises a heat dissipation hole.

11. An electronic device, comprising: a storage device, comprising: a first side surface, having a first hole; and two second side surfaces opposite to each other, respectively adjacent to the first side surface, and each of the second side surfaces has a second hole; and a fixing assembly, comprising: a supporting body, comprising a base plate and a side plate connected to each other, wherein the base plate is adapted to carry the storage device, and the side plate has a first positioning column extending in a first axial direction; and two moving components, movably disposed on two sides of the base plate in a second axial direction, wherein an accommodating space is formed between the supporting body and the two moving components, the two moving components respectively comprise two second positioning columns extending in the second axial direction, and the first positioning column and the two second positioning columns protrude towards the accommodating space; when the storage device is disposed in the accommodating space, the first positioning column inserted into the first hole, and the two second positioning columns adapted to be inserted into the two second holes.